

Syn2bANI: chain-restricted maximum-likelihood ANI on fixed restriction-site anchors

a in-silico digestion

— Bcgl — Alfi — Aloi — Fall

genome A

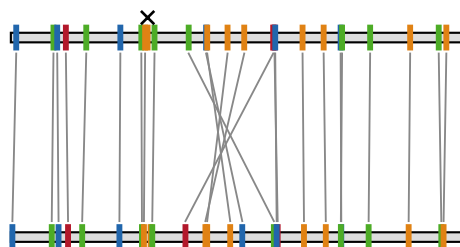


genome B



positioned Type IIB tags (27–32 bp)

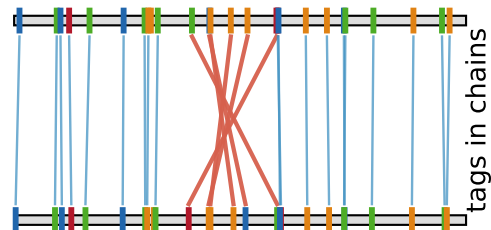
b anchoring (≤ 2 mismatches)



*shared tags $\rightarrow$ anchors
(pigeonhole seeding, no scan)*

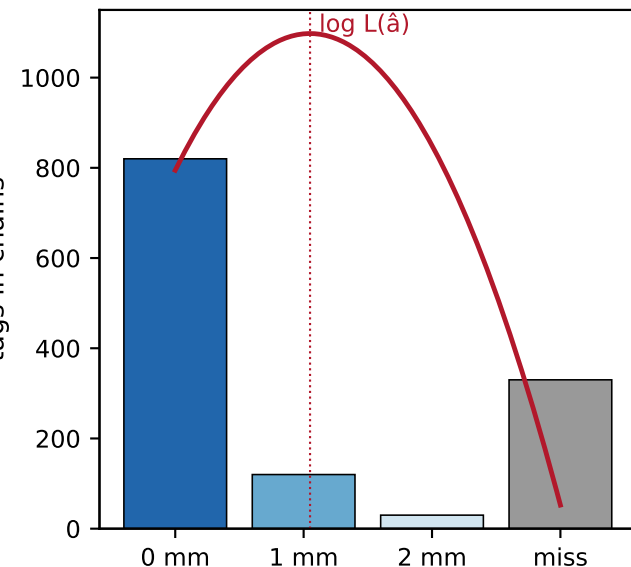
c collinear chaining (2-pass)

— + chains — (inverted) block



*gap-penalized DP;
adaptive chain-break test*

d chain-restricted MLE



- ANI $\pm$ standard error (gated gamma / uniform MLE)
- aligned fraction af_query / af_reference
- synteny_score, synteny_blocks, breakpoint_count
- SV calls: inversions, indels ≥ 1 kb, translocations
- reliability flag: ok / INCONSISTENT / BELOW_DETECTION
- ani_upper95 when no point estimate is responsible

*one sketch, one pass:
~8 ms per pair, 58 MB peak RSS
(vs 8–10 s per pair for dnadiff)*